Vector of vectors in C++

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1. can I do this with vector in c++:
   pr[0].push_back(0);
   pr[1].push_back(0);

Yes ✅ — you can absolutely do that in C++, **if `pr`** is a **vector of vectors** (a 2D vector).

Here's what that means:

✅ Example where it works:

```cpp
#include <bits/stdc++.h>
using namespace std;

int main() {
    vector<vector<int>> pr(2);  // creates 2 empty inner vectors: pr[0] and pr[1]

    pr[0].push_back(0);  // works
    pr[1].push_back(0);  // works

    cout << pr[0][0] << " " << pr[1][0] << endl; // prints: 0 0
}
```

⚠️ When it won't work:

If you declared `pr` like this:

```cpp
vector<int> pr;
```

then `pr[0]` and `pr[1]` don't exist — because it's a **1D vector**, not a vector of vectors.
Trying to do `pr[0].push_back(0);` will cause a **compile-time error**, since `pr[0]` would be an `int`, not a vector.

✅ Rule of thumb:

- Use `vector<vector<int>> pr(n);` if you want multiple inner lists (like a table or matrix).
- Then you can safely do `pr[i].push_back(value);` for any valid index `i`.

Would you like me to show a visual of how memory looks for `vector<vector<int>> pr(2);` after these operations?

2. SieveOfEratosthenisPrac.cpp
File

what is wrong in this code?

Good question — I looked at your code, and there are **two main logic errors** in your implementation of the **Sieve of Eratosthenes**.

Here's the relevant part with problems:

```cpp
for(int i = 3; i<=n; i=i+2)  // Odd
{
    if(pr[i]==1)
    {
        for(int j=i*i; j<=n; j=i+i)
        {
            pr[i]=0;           // Not Prime ❌ wrong
        }
    }
}
```

Let's break it down:

❌ 1. Wrong loop increment inside the inner loop

You wrote:

```cpp
for(int j=i*i; j<=n; j=i+i)
```

That means **`j` never changes** (it's always `i + i`), so you end up in an **infinite loop** or only affect one element.

✅ It should be:

```cpp
for (int j = i * i; j <= n; j += i)
```

This ensures every multiple of `i` is marked.

❌ 2. Wrong index being marked as not prime

Inside the inner loop, you're marking:

```cpp
pr[i] = 0;   // ❌ marking i instead of j
```

You need to mark the multiples (**j**), not the prime itself.

✅ It should be:

```cpp
pr[j] = 0;
```

✅ Corrected version:

```cpp
#include <bits/stdc++.h>
using namespace std;

int main() {
    int n = 100;
    vector<int> pr(n + 1, 1);   // All indices marked as prime initially
    pr[0] = pr[1] = 0;          // 0 and 1 are not prime

    // Mark even numbers > 2 as not prime
    for (int i = 4; i <= n; i += 2)
        pr[i] = 0;

    // Process odd numbers
    for (int i = 3; i * i <= n; i += 2) {
        if (pr[i] == 1) {
            for (int j = i * i; j <= n; j += i)
                pr[j] = 0;      // Mark multiples of i
        }
    }

    // Print primes
    for (int i = 2; i <= n; i++) {
        if (pr[i] == 1)
            cout << i << " ";
    }

    return 0;
}
```

✅ Output:

```
2  3  5  7  11  13  17  19  23  29  31  37  41  43  47  53  59  61  67  71  73  79  83  89  97
```

Would you like me to show a **step-by-step dry run** (like how it marks 2, 3, 5, etc.) to help visualize how the sieve works?